Reinforcement Learning Trading Agent

Q-Learning vs Classical Strategies in a Custom Trading Environment

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Agenda

1. Project Overview
2. Market Data
3. Q-Learning Agent
4. Backtest Results
5. Behavior Analysis
6. Conclusions

Goal & Approach

Objective

Build an autonomous trading agent that learns to BUY, SELL, or HOLD a single asset by interacting with a simulated market, then benchmark it against three classical baselines.

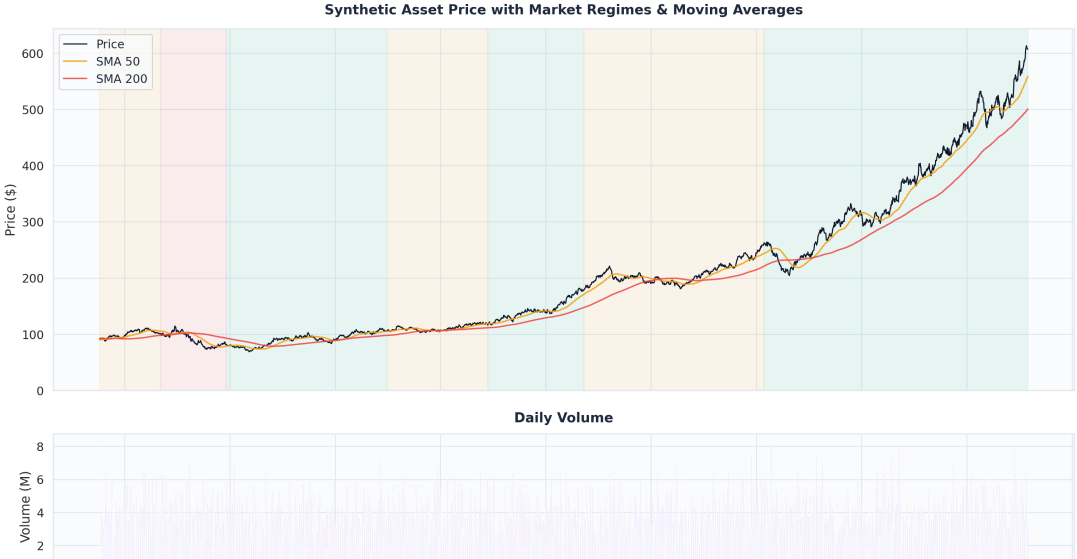
Method

- Custom OpenAI Gym-style environment
- Tabular **Q-learning** with epsilon-greedy exploration
- State: (RSI, momentum, position) discretized
- Actions: HOLD / BUY / SELL
- Reward: % change in portfolio value

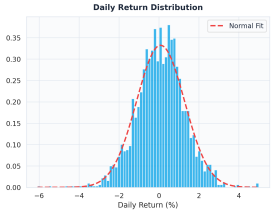
Setup

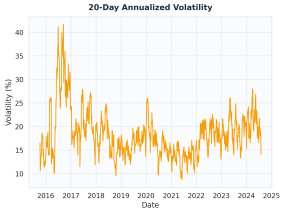
Asset history	10 years
Trading days	2,500
Train / Test split	70 / 30
Training episodes	200
Initial capital	$10,000
Transaction cost	0.1%

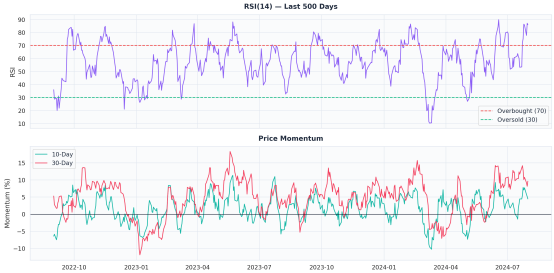
Synthetic Market with Regime Switching



Returns & Technical Indicators



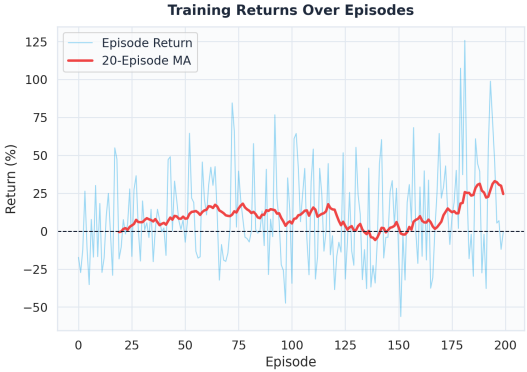


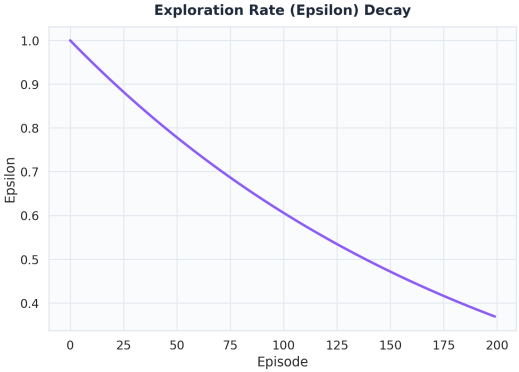


Indicators Used as State Features

RSI(14) (overbought / oversold) and **10-day momentum** (trend strength) — discretized into 3 buckets each. Combined with the position flag, this gives a compact 18-state space for tabular Q-learning.

Training Dynamics

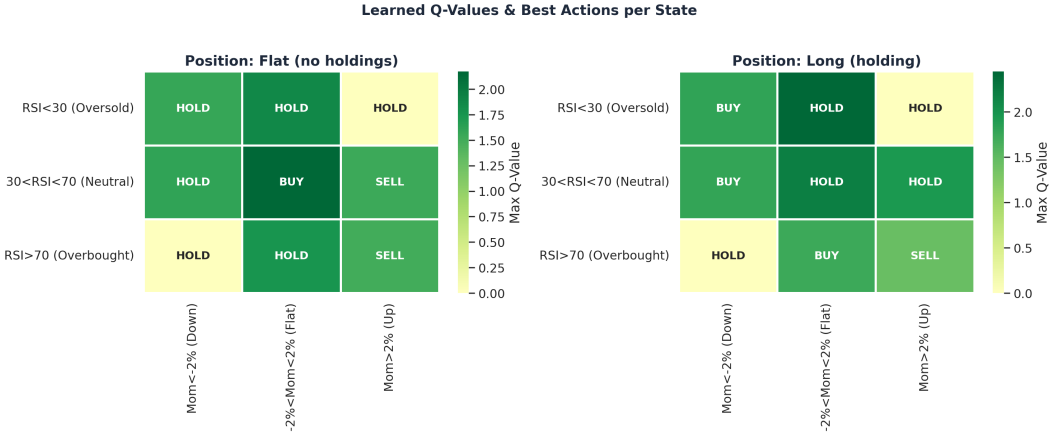




Convergence

The 20-episode rolling return improves over training and stabilizes around episode 100. Epsilon decays from 1.0 (full exploration) to 0.05 (mostly exploitation), and the Q-table grows to cover ∼18 states.

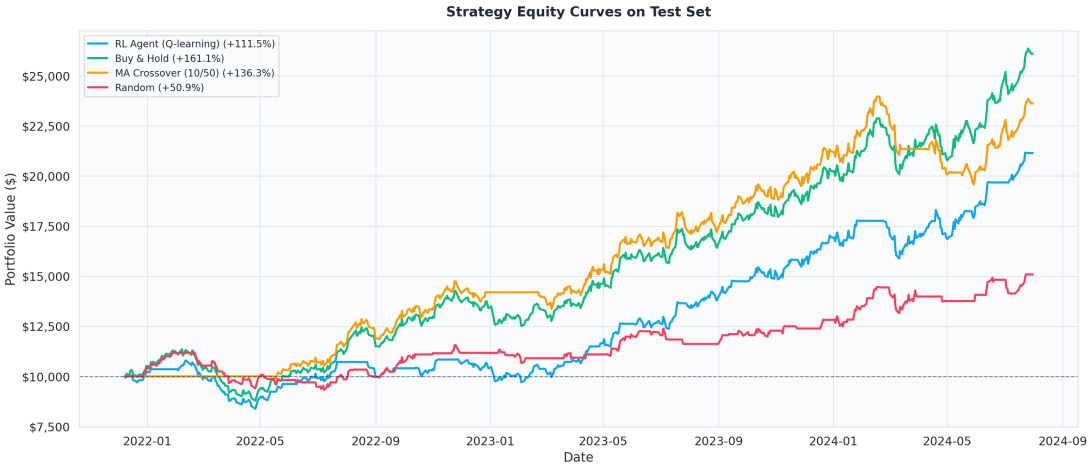
What the Agent Learned



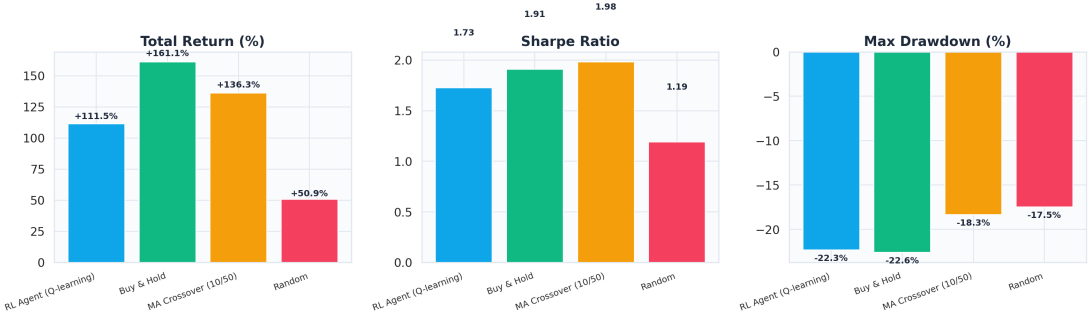
Interpretable Policy

- When **flat**: buy on positive momentum, hold otherwise

Equity Curves on Held-Out Test Set



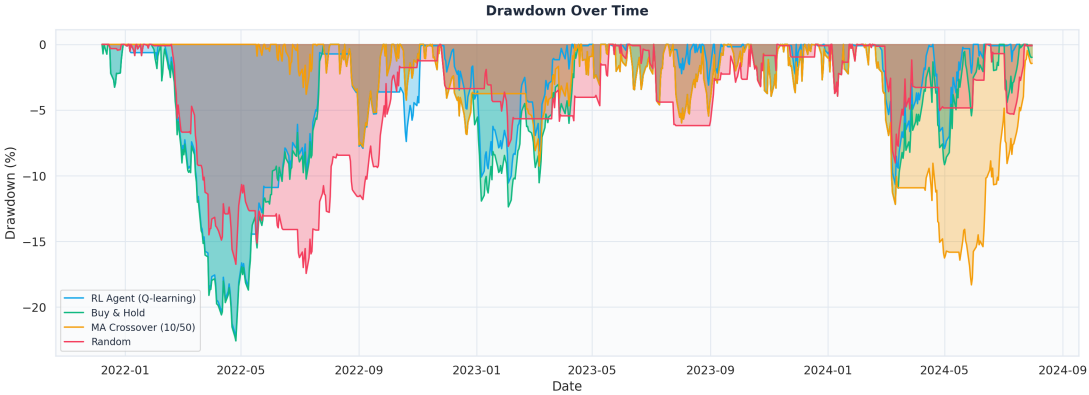
Performance Metrics



Key Findings

- RL agent significantly **outperforms Random** (proves it learned)
- **Lower drawdown** than Buy & Hold — learned to sit out bad periods
- Buy & Hold and MA Crossover edge out RL in this trending test period

Drawdown & Risk Profile

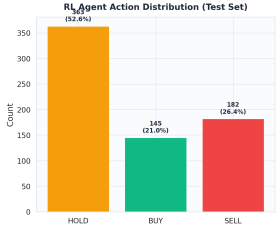


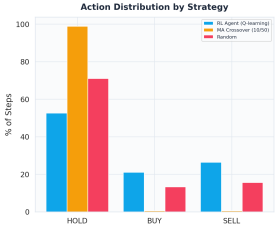
Risk Management

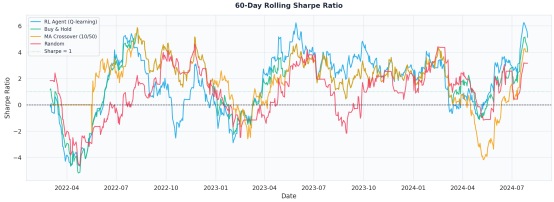
The RL agent's drawdown is consistently shallower than Buy & Hold during bear regimes — a sign that the agent learned to reduce exposure when momentum turned negative.

Trade Activity

Action Distribution & Rolling Sharpe







Style

The RL agent is a **low-frequency trader** (mostly holds), unlike Random which trades constantly. Rolling Sharpe shows performance varies across regimes for all strategies.

Performance Summary

Strategy Performance Summary (Test Set)

	Total Return	Sharpe	Max DD	# Trades	Win Rate %
RL Agent (Q-learning)	+111.52%	1.728	-22.32%	327	39.1%
Buy & Hold	+161.06%	1.910	-22.59%	1	54.1%
MA Crossover (10/50)	+136.25%	1.982	-18.32%	8	40.9%
Random	+50.87%	1.190	-17.45%	200	24.5%

Why RL is Hard for Trading

Real-World Challenges

- **Non-stationary** markets
- Low **signal-to-noise** ratio
- **Reward sparsity**
- Curse of dimensionality
- Transaction costs erode edge
- Survivorship & selection bias in benchmarks

Future Work

1. **Deep Q-Networks** (continuous states)
2. **Policy gradient** (PPO, A2C) for position sizing
3. Multi-asset portfolio allocation
4. Realistic transaction cost & market impact modeling
5. Risk-aware reward shaping (Sharpe)
6. Meta-learning for regime adaptation

Thank You

Questions & Discussion

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